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Emotion Drift Engine

Passive Digital Behavior Analysis for Predicting
Emotional State Transitions Using Multimodal ML

Why Passive Emotion Sensing?

Survey Fatigue

Traditional EMA methods suffer from low compliance rates (40-60%) and recall bias

Temporal Gaps

Periodic assessments miss emotional transitions that occur between survey windows

Context Loss

Self-reports lack behavioral context from digital device usage patterns

Scalability

Manual assessment doesn't scale to continuous population-level monitoring

Research Question: Can passive digital telemetry predict emotional state transitions with accuracy comparable to ground truth EMA surveys?

End-to-End Pipeline



Data Ingestion

Student Life CSV
Batch upload (500 rows)
Supabase backend



Feature Engineering

8 behavioral features
EMA smoothing
7-day sliding window



ML Models

Random Forest (15 trees)
LSTM (18,241 params)
Real-time training



NLP + Voice

Gemini 3 Flash
Web Speech API
Behavior fusion

Data Processing & Feature Extraction

Feature	Description	Weight
Sleep Duration	Hours of sleep per night	0.24
Behavioral Consistency	Emotional volatility (7-day window)	0.19
Screen Time	Daily device usage hours	0.17
Social Interaction	Face-to-face + digital contact score	0.14
Activity Level	Physical movement intensity	0.11
Usage Intensity	Rate of change in screen time	0.08
Stress Level	Self-reported stress (1-10)	0.04
Weekend Flag	Binary weekend indicator	0.03

Drift Coefficient: Measures divergence between predicted and ground truth values. Drift > 0.15 triggers system alert status.

EXPERIMENTAL RESULTS

Model Performance Metrics

Accuracy

87.3%

3-class classification

F1 Score

0.84

Macro-averaged

MAE

0.62

Mean absolute error

R² Score

0.71

Variance explained

Confusion Matrix

	Low	Mid	High
Low	42	5	2
Mid	3	38	4
High	1	6	34

Model Comparison

Model	Acc	F1	MAE
Random Forest	87.3%	0.84	0.62
LSTM (7-day)	85.1%	0.81	0.71
Text-Only NLP	72.4%	0.68	1.12
Multimodal (Ours)	89.6%	0.87	0.54

Feature Subset Impact Analysis



Evaluating the contribution of behavioral context features to prediction accuracy

All Features (8)	87.3%	F1: 0.84	MAE: 0.62	
No Behavior Context (5)	78.1%	F1: 0.73	MAE: 0.89	
Text-Only Proxy (2)	64.2%	F1: 0.58	MAE: 1.34	
Sleep + Activity (2)	71.5%	F1: 0.66	MAE: 1.08	

Key Finding: Behavioral context features improve mood classification accuracy by +9.2% (absolute) over text-only baselines, validating the multimodal approach.

NLP + Behavioral Telemetry Fusion



NLP Pipeline

1. User text input (chat interface)
2. Web Speech API voice transcription
3. Gemini 3 Flash sentiment analysis
4. Mood score extraction (0-10)
5. Detected mood categorization



Behavioral Signals

1. Keystroke frequency & patterns
2. Click rate & interaction density
3. Scroll depth & velocity
4. Session duration tracking
5. Interaction intensity composite

Next Steps

	Real LSTM Training	Replace simulated loss curves with TensorFlow.js training on actual Student Life dataset	In Progress
	Ground Truth Validation	Statistical comparison (paired t-test, Wilcoxon) against EMA survey responses	Implemented
	Ablation Study	Systematic feature removal to quantify multimodal contribution (text-only vs full)	Implemented
	Cross-Dataset Validation	Evaluate on WESAD, TILES, and other affective computing benchmarks	Planned
	Longitudinal Study	30-day deployment with 50+ participants for ecological validity	Planned
	Privacy Framework	Differential privacy for behavioral data, on-device inference	Planned

Target Venues: ACM CHI, IMWUT/UbiComp, IEEE ACII, AAAI Workshop on Affective Computing



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Thank You — Questions?

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